The Riemann Integral

Bartle & Sherbert §7.1 (7.1.1–7.1.7), §7.2 (7.2.1–7.2.13), §7.3 (7.3.1–7.3.18) · FYUGP pattern (April 2026), doubled

HOW THIS PAPER WORKS

A real FYUGP paper has 10 short (3 marks, ceiling 24), 8 paragraph (6 marks, ceiling 36) and 2 essay (10 marks, answer one) questions. This paper has **twice as many** (20 + 16 + 4 = 40 questions, 196 marks), all from Module 3. Each of the module's 32 atomic concepts (T01–T32) and 8 core application types is the main point of exactly one question.

Time	4 hours (or two sittings of 2 hours)
Maximum	140 marks (A ceiling 48, B ceiling 72, C answer any two = 20)

PYQ MTS-25 Q6 = MTS 6B 10, March 2025, question 6 (old CBCSS course, same textbook). "also" = another paper asked the same thing. "adapted" = a 1-mark MCQ turned into a 3-mark or 6-mark question with full justification.

Section A · Short answer

All questions can be answered. Each carries 3 marks. Ceiling 48.

1. Define a partition of [a,b] and the norm of a partition. Find the norms of $P_1 = (0, 1, 2, 4), P_2 = (0, 2, 3, 4)$, and $P_3 = (0, 1, 1.5, 2, 3.4, 4)$ on [0,4]. Which of $Q_1 = (2, 4, 6, 8, 9), Q_2 = (2, 6, 9), Q_3 = (2, 5, 7, 9)$ on [2,9] has the largest norm?

`PYQ MTS-26 Q4` `also MAT-22 Q3, MAT-21 Q5` `MAT-22 MCQ 16 · adapted`

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2. Define a tagged partition and the Riemann sum S(f;\dot P) of a function f : $[a, b] \to \mathbb{R}$. For f(x) $= x^2$ on [0,4] and P = (0,1,2,4), compute S(f;\dot P) with tags at left endpoints and at right endpoints.

`PYQ MTS-22 Q4`

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3. Let f : $[a, b] \to \mathbb{R}$. State the ε–δ_ε definition of "f is Riemann integrable on [a,b]" (including the notation $\mathcal{R}[a, b]$ and $\int_a^b$ f). Why is $\int_a^b f$ not an ordinary function limit as $\lVert P\rVert \to 0$?

`PYQ MAT-21 MCQ 9 · adapted`

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4. Prove that if $f \in \mathcal{R}[a, b]$, then the value of the integral $\int_a^b f$ is uniquely determined.

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5. Prove directly from Definition 7.1.1 that every constant function f(x) = k on [a,b] is Riemann integrable on [a,b], and find $\int_a^b k$.

PYQ MTS-25 Q4

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6. If $g \in \mathcal{R}[a,b]$ and f(x) = g(x) except for a finite number of points in [a,b], prove that $f \in \mathcal{R}[a,b]$ and $\int_a^b f = \int_a^b g$ (proving the case of one exceptional point suffices). Hence find $\int_0^2 h$ where h(x) = 2 on [0,1), h(1) = 3, and h(x) = 1 on (1,2]. What happens if f(x) = 0 except at finitely many points, or if f(x) = g(x) off the infinite set $\{1/n : n \in \mathbb{N}\}$?

PYQ MAT-22 MCQ 18 · adapted

also MAT-21 MCQ 13, MTS-22 MCQ 7 · adapted

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7. State and prove the Boundedness Theorem for Riemann integrable functions. Deduce that H(1/k) = k ($k \in \mathbb{N}$), H(x) = 0 elsewhere on [0,1], is not in $\mathcal{R}[0,1]$. Does $f \in \mathcal{R}[a,b]$ imply f is continuous or monotone? Is every bounded function integrable?

PYQ MAT-22 MCQ 15 · adapted

also MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9 · adapted

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8. If J is a subinterval of [a,b] with endpoints c < d and $\varphi_J(x) = 1$ on J, 0 elsewhere on [a,b], prove that $\varphi_J \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_J = d - c$. Hence evaluate $\int_0^4 g$ where g(x) = 3 for $0 \le x \le 2$ and g(x) = 2 for $2 < x \le 4$.

PYQ MAT-21 MCQ 10 · adapted

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9. For $f \in \mathcal{R}[a,b]$ and $\alpha < \beta$ in [a,b], define $\int_\beta^\alpha f$ and $\int_\alpha^\alpha f$. State the additivity formula for arbitrary $\alpha, \beta, \gamma \in [a,b]$ and verify it for f(x) = x with $\alpha = 3, \beta = 1, \gamma = 0$.

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10. Explain why the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1) cannot be applied to H(x) $= 2\sqrt{x}$ on [0,b], nor to K(x) $= x^2 \cos(1/x^2)$ (x $\in (0,1]$), K(0) = 0, on [0,1], even though H is continuous on [0,b] and K is differentiable at every point of [0,1].

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11. Define the indefinite integral F of $f \in \mathcal{R}[a,b] with base point a. Prove that if |f(x)| ≤ M on $[a,b]$, then $|F(z) - F(w)| \le M|z - w|$ for all z, w $\in$ [a,b]$, and deduce that F is continuous on [a,b].

PYQ MAT-21 Q6

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12. Find the indefinite integral F(x) $= \int_{-1}^x \operatorname{sgn} of f(x) = \operatorname{sgn} x$ on [-1,1] with basepoint -1. Is F an antiderivative of f on [-1,1]? What about H(x) $= \int_0^x h$ for Thomae's function h on [0,1]?

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13. Define a null set in $\mathbb{R}$. Prove that the set $\mathbb{Q} \cap [0,1]$ of rational numbers in [0,1] is a null set. State whether subsets and unions of null sets are null, and whether every null set is countable.

PYQ MTS-25 Q7 **also MTS-26 Q7**
MTS-22 MCQ 10 · adapted

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14. Using the First Form of the Fundamental Theorem of Calculus: (a) state $\int_a^b f(x)\,dx$ when f is continuous and F is an antiderivative of f on [a,b]; (b) evaluate $\int_{-2}^{3} g(x)\,dx$ where g(x) = x for $|x| \geq 1$ and g(x) = -x for $|x| < 1$, using G(x) $= \frac{1}{2}|x^2 - 1|$; (c) show that $\int_a^b |x|\,dx =$ B(b) - B(a) where B(x) $= -\frac{1}{2}x^2$ (x < 0), $\frac{1}{2}x^2$ (x $\geq$ 0), stating the exceptional set E in (b) and (c).

PYQ MAT-22 MCQ 27, 28, 29 · adapted
also MAT-21 MCQ 19, 20 · adapted

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15. Evaluate the integral $\displaystyle\int_0^1 x^2(1 - \sqrt x)\,dx$, justifying each hypothesis of the Fundamental Theorem of Calculus that you use.

PYQ MAT-22 Q35

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16. (a) If f is continuous on [-1,1], use the Substitution Theorem to show that $\int_0^{\pi/2} f(\cos x)\,dx = \int_0^{\pi/2} f(\sin x)\,dx = \tfrac12\int_0^\pi f(\sin x)\,dx$ ($and is not in general equal to \int_0^\pi f(\sin x)\,dx$). (b) If f \in \mathcal{R}[a,b] and $g(y) := f(y-c)$ on $[a+c, b+c]$, show that $\int_{a+c}^{b+c} g = \int_a^b f$.

PYQ MAT-22 MCQ 19, 26 · adapted

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17. If $f : [0,1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0,1]$, prove that $f \in \mathcal{R}[0,1]$ and f(x) = 0 for all $x \in [0,1]$.

PYQ MAT-22 MCQ 30 · adapted

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18. Let f be continuous on [a,b] with f(x) ≥ 0 for all $x \in [a,b]$. If $\int_a^b f = 0$, prove that f(x) = 0 for all $x \in [a,b]$. Show by an example that the continuity of f cannot be dropped.

PYQ MAT-22 MCQ 23 · adapted

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19. State the rule for differentiating G(x) $= \int_{u(x)}^{v(x)} f(t)\,dt$ when f is continuous and u, v are differentiable. Hence find F'(x) on [0,1] for: (a) $F(x) = \int_0^{x^2} (1+t^3)^{-1}\,dt$; (b) $F(x) = \int_{x^2}^{x} \sqrt{1+t^2}\,dt$.

PYQ MTS-22 Q14

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20. (Mean Value Theorem for Integrals) If f : $[a,b] \to \mathbb{R} $is continuous on $[a, b] (a < b)$, prove that there exists c \in [a,b] such that \int_a^b f =$ f(c)(b - a). Does this hold for step functions?

Section B · Paragraph / problem
All questions can be answered. Each carries 6 marks. Ceiling 72.

21. Suppose f and g are in $\mathcal{R}[a,b]$. Prove that: $(a) f+g \in \mathcal{R}[a,b]$ and $\int_a^b$ (f + g) $= \int_a^b f + \int_a^b g$; (b) if f(x) $\leq$ g(x) for all $x \in [a,b]$, then $\int_a^b f \leq \int_a^b g$ (and show why strict inequality < need not hold).

PYQ MAT-21 Q29 · also MTS-22 Q5, MTS-25 Q5 · MAT-22 MCQ 14 · adapted

22. Define Thomae's function h on [0,1]. Prove directly from Definition 7.1.1 that $h \in \mathcal{R}[0,1]$ and $\int_0^1 h = 0$. Explicitly list all elements of the set $E = \{x \in [0,1] : h(x) \geq 1/5\}$.

23. State and prove the Cauchy Criterion for the Riemann integrability of f : $[a,b] \to \mathbb{R}$.

PYQ MAT-21 Q8

24. Prove using the Cauchy Criterion that Dirichlet's function (f(x) = 1 on $\mathbb{Q} \cap [0,1]$, 0 on $[0,1] \setminus \mathbb{Q}$) is not Riemann integrable on [0,1]. Also prove that h(x) = x + 1 on $\mathbb{Q} \cap [0,1]$, h(x) = 0 on $[0,1] \setminus \mathbb{Q}$, is not in $\mathcal{R}[0,1]$.

PYQ MAT-22 Q4 · also MAT-22 MCQ 17, MAT-21 MCQ 12, 16 · adapted

25. State and prove the Squeeze Theorem for Riemann integrals. Use it with step functions on an n-part equal partition to prove that h(x) = x belongs to $\mathcal{R}[0,1]$ with $\int_0^1 x\,dx = \frac{1}{2}$.

PYQ MTS-26 Q18 **also MTS-22 Q6, MAT-21 Q7**

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26. Prove that every step function φ : $[a,b] \to \mathbb{R} $is Riemann integrable, and derive the formula $\int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j)$. Use it to evaluate $\int_0^3 g$ where g(x) = 2 on [0,1] and g(x) = 3 on (1,3].

PYQ MTS-26 Q5 **also MAT-22 Q16**

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27. If $f \in \mathcal{R}[a,b]$ and $[c,d] \subseteq [a,b]$, prove that the restriction of f to $[c,d]$ belongs to $\mathcal{R}[c,d]$. Deduce by mathematical induction that if $a = c_0 < c_1 < \cdots < c_m = b$, then $\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f$.

PYQ MAT-22 Q37b

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28. Let f be bounded by M on [a,b] and suppose the restriction of f to [c,b] belongs to $\mathcal{R}[c,b]$ for every $c \in (a,b)$. Prove that $f \in \mathcal{R}[a,b]$ and $\lim_{c\to a^+} \int_c^b f = \int_a^b f$. Deduce that $g(x) = \sin(1/x)$ $(x \in (0,1])$, $g(0) = 0$, belongs to $\mathcal{R}[0,1]$, and show by an example that the boundedness of f cannot be dropped.

PYQ MAT-22 MCQ 24, 25 · adapted

also MAT-21 MCQ 18 · adapted

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29. State and prove the First Form of the Fundamental Theorem of Calculus (prove the case $E = \{a, b\}$ in full and show how the general finite set E follows).

PYQ MTS-26 Q8

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30. State and prove the Substitution Theorem for Riemann integrals. Use it to evaluate $\int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt, \int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt, \int_0^1 t\sqrt{1+t^2}\,dt, \int_0^2 t^2(1+t^3)^{-1/2}\,dt$, and \int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt. Finally, explain why the Substitution Theorem with $\varphi(t) = \sqrt t $cannotbeappliedto$ \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt, $andevaluate$ \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt$ using the First Form of the Fundamental Theorem.

PYQ MTS-25 Q6 **also MAT-22 Q18, MAT-21 Q19**

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31. State Lebesgue's Integrability Criterion. Use it to determine the Riemann integrability on [0,1] of: (a) any step function; (b) any monotone function; (c) G(1/n) = 1/n ($n \in \mathbb{N}$), G(x) = 0 elsewhere; (d) Dirichlet's function; (e) Thomae's function.

PYQ MAT-22 Q6

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32. State and prove the Composition Theorem for Riemann integrable functions. Show by an explicit counterexample that the continuity of the outer function φ cannot be dropped.

PYQ MAT-21 Q4

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33. If $f \in \mathcal{R}[a,b]$ with $\lvert f(x)\rvert \le M on[a,b], $prove that$\lvert f\rvert \in \mathcal{R}[a,b] and\left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert \le $M(b-a). $Show by an example that$\lvert f\rvert \in \mathcal{R}[a,b]$ does not imply $f \in \mathcal{R}[a,b]$.

PYQ MTS-25 Q8 **also MTS-22 Q7, MAT-21 Q20**

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34. State and prove the Product Theorem: if $f, g \in \mathcal{R}[a,b], $then fg$\in \mathcal{R}[a,b]. $Give an example to show that$\int_a^b fg\ne \left(\int_a^b f\right)\left(\int_a^b g\right)$ in general.

PYQ MTS-26 Q6 **also MAT-22 Q17, MAT-21 Q18**

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35. State and prove the Integration by Parts Theorem for Riemann integrals. Use it to evaluate $\int_0^1 xe^x\,dx$.

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36. State and prove Taylor's Theorem with the Integral Form of the Remainder. Verify the theorem for f(x) $= e^x$ on [0,1] (a = 0, b = 1) with n = 1 and n = 2.

PYQ MTS-25 Q19 **also MAT-22 Q32**

Section C · Essay

Answer any two. Each carries 10 marks.

37. Prove that if f : $[a, b] \to \mathbb{R}$ is continuous on [a,b], then $f \in \mathcal{R}[a, b]$. State clearly where each hypothesis is used in the proof. Is every function in $\mathcal{R}[a, b]$ continuous on [a,b]? Is every function in $\mathcal{R}[a, b]$ uniformly continuous on [a,b]? Justify with examples.

PYQ MTS-25 Q18 **also MAT-21 Q30, MTS-22 Q19** **MAT-22 MCQ 21, MAT-21 MCQ 15 · adapted**

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38. Prove that if f : $[a, b] \to \mathbb{R}$ is monotone on [a,b], then $f \in \mathcal{R}[a, b]$. Illustrate the proof by computing the lower and upper step-function integrals $\int_0^1 \alpha$ and $\int_0^1 \omega$ for f(x) $= x^2$ on [0,1] with n = 4 equal subintervals. Is every function in $\mathcal{R}[a, b]$ monotone on [a,b]?

PYQ MTS-26 Q19 **also MAT-22 Q30** **MAT-22 MCQ 22 · adapted**

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39. State and prove the Additivity Theorem: let f : $[a, b] \to \mathbb{R}$ and $c \in (a, b)$; prove that $f \in \mathcal{R}[a, b]$ if and only if its restrictions to [a,c] and [c,b] are both in $\mathcal{R}$, and in this case $\int_a^b f = \int_a^c f + \int_c^b f$.

PYQ MTS-26 Q25 **also MAT-22 Q37a, MTS-22 Q25**

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40. State and prove the Second Form of the Fundamental Theorem of Calculus (both the pointwise form at $c \in [a, b]$ and the global consequence when f is continuous on [a,b]). Give an example of a function $f \in \mathcal{R}[-1,1]$ whose indefinite integral fails to be differentiable at 0.

PYQ MTS-25 Q25 **also MAT-21 Q37a, MAT-22 Q33**

Coverage

TOPIC	QUESTIONS
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T02 Tagged partitions, Riemann sums	2
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T04 Uniqueness of the integral (7.1.2)	4
T05 Integrals from the definition (7.1.4)	5
T06 Finite exceptional set (7.1.3)	6
T07 Linearity and monotonicity (7.1.5)	21
T08 Boundedness Theorem (7.1.6)	7
T09 Thomae's function (7.1.7)	22
T10 Cauchy Criterion (7.2.1)	23
T11 Dirichlet and relatives $\notin \mathcal{R}$(7.2.2b)	24
T12 Squeeze Theorem (7.2.3)	25
T13 Elementary step functions φ_J(7.2.4)	8
T14 Step functions $\in \mathcal{R}[a,b]$ (7.2.5)$< /td >< td > 26 < /td >< /tr >< tr >< td > T15 Continuous$\Rightarrow $\mathcal{R}[a,b]$ (7.2.7) & applications	18, 20, 37
T16 Monotone $\Rightarrow \mathcal{R}[a,b]$ (7.2.8)	38
T17 Additivity Theorem (7.2.9)	39
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T21 Examples and limits of FTC 1 (7.3.2)	10, 14, 15

T22 Indefinite integral is Lipschitz (7.3.3, 7.3.4)	11
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T26 Null sets & $\mathbb{Q} \cap [0,1]$ (7.3.10, 7.3.11)	13
T27 Lebesgue's Integrability Criterion (7.3.12, 7.3.13)	31
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T29 Absolute value $\lvert f \rvert \in \mathcal{R}[a,b] (7.3.15)$	33
T30 Product Theorem fg $\in \mathcal{R}[a,b] (7.3.16)$	34
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T32 Taylor's Theorem with Integral Remainder (7.3.18)	36

All 74 past questions appear (plus the 3 additional MTS-22 descriptive rows in Part 8: 77 past questions in total — 46 written, 31 MCQs).